From Narrative Incident Reports to Encounter-Risk Maps: A Knowledge Graph Pipeline for Human–Bear Conflict in Japan

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Abstract

Bear incident records in Japan are fragmented across heterogeneous sources and are often reported as narrative text rather than structured data. We present a reproducible pipeline that aggregates thousands of reports from government websites and citizen-report platforms, extracts structured attributes into a controlled JSON schema using a large language model (LLM), and loads the results into a knowledge graph (KG) for querying and analysis. To screen for likely repeat encounters, we cluster reports based on spatio-temporal proximity, augmented with reported bear size and group composition un sin Neo4j . Finally, we use the resulting SDM outputs as risk-guidance layers, presented through a web interface alongside mapped bear occurrence points, attack-type categories, and clusters of likely related encounters derived from similarity analysis, providing an integrated situational overview for public-safety planning.

Keywords

LLM, Metadata, Knowledge Graph

1. Introduction

Human–bear conflicts have increased in Japan, and recent reporting describes a sharp 2025 spike, including more than 100 attacks and a record 12 deaths.[1] Operationally, stakeholders need to identify where incidents recur, when risk peaks, and how management actions relate to subsequent reports. Yet incident records are scattered across heterogeneous sources and formats, and key details (bear traits, setting, and response) are embedded in narrative descriptions.

Operational planning also requires spatial generalization. Incident dashboards and heatmaps are reactive summaries of where reports occurred; they do not provide spatially continuous estimates of where encounters are likely under similar environmental conditions. Species distribution models (SDMs) address this gap by relating occurrence locations to environmental covariates (elevation, land cover, distance to forest edge, etc.) to produce mapped suitability or encounter-risk proxies [1, 2]. Unlike kernel density maps of past reports, SDMs can extrapolate risk to unreported but environmentally similar areas and support scenario comparisons. Such models have been used to map human–wildlife conflict risk hotspots and guide mitigation strategies [3, 4, 5].

Knowledge graphs (KGs) provide a practical representation for linking incidents, places, and actions to support queries spanning time, space, and context [6, 7]. Large language models (LLMs) make it feasible to convert narrative reports into structured fields at scale when guided by a controlled schema [8, 9]. We build on these capabilities to integrate national-scale reporting and connect incident-derived observations to ecological modeling workflows.

2. Methods

Data aggregation and normalization. We ingest incident reports from multiple streams (e.g., municipal advisories/summaries and citizen reporting platforms) and normalize them into a common

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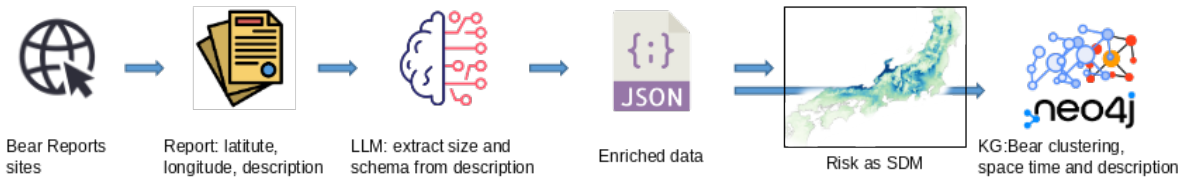
[1]Reuters. *Japan deploys troops to combat deadly wave of bear attacks.* November 5, 2025. https://www.reuters.com/business/environment/japan-sends-troops-combat-deadly-wave-bear-attacks-2025-11-05/

record format while preserving provenance (source identifiers/URLs and raw text). Each record includes at least a date (or date range) and a place reference; coordinates are stored when available or derived via geocoding with an associated precision indicator.

Controlled schema and LLM extraction. We use a controlled JSON schema with constrained value lists for: (i) incident type (e.g., sighting, crop raiding, building entry, attack), (ii) bear traits (species when stated, size cues, and group composition such as single vs. sow-with-cubs), (iii) environmental setting (e.g., residential edge, farmland), and (iv) response actions (warnings, patrols, trapping, capture/relocation, lethal control). An LLM extract such metadata from te narrative text into these fields.

Knowledge graph in Neo4j. Each report becomes an node with temporal and spatial properties. Reusable nodes represent shared entities (types, traits, contexts, responses, and locations) and connect to incidents. To identify reports plausibly associated with the same bear (or bear group), we build a similarity graph over incidents: edges connect reports that are close in space and time *and* consistent in reported size and group composition when these attributes are available. We then extract connected components using Neo4j as candidate repeat-event clusters. This linkage is a screening heuristic for repeated local conflict; it does not assert definitive individual identity.

From incident reports to SDM-ready occurrences. Using the latitude and longitude fields extracted into the controlled JSON schema, we assemble georeferenced presence points for modeling. And train two separate SDMs[2]: one using black bear records and another using brown bear records.



3. Use Cases, Limitations, and Future Work

The KG supports queries that combine place, time, incident type, and response actions (e.g., identifying areas with repeated building-entry incidents in late autumn and summarizing whether responses escalate from warnings to trapping). The clustering step strengthens these summaries by grouping likely repeat encounters, enabling analysts to focus on recurring problems rather than isolated reports. In addition, SDMs trained on the derived occurrence set can produce seasonal encounter-risk surfaces that complement the KG: the graph explains *how* conflicts occurred (context and response), while SDM outputs indicate *where* similar conditions exist and where preventive actions may be prioritized.

Limitations follow from heterogeneous reporting, missing traits/species, and location uncertainty, as well as observation bias toward human-used landscapes. Future work extends the KG with geospatial ontologies and environmental variables and evaluates bias-aware SDM designs that better separate encounter likelihood from reporting effort [10].

Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

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